L5

One source of truth for features

Same model. Same features. DIFFERENT code paths — offline in a notebook, online in a microservice. Subtle rounding, missing-value handling, time-window definition. Welcome to training-serving skew — the silent killer.

Have you ever shipped a model whose production score surprised you?

WHAT'S INSIDE

01 Training-serving skew

02 What a feature store does

03 Feature store landscape 2026

04 DVC — Git for data

05 Analogy

WHY THIS LESSON MATTERS

Offline model: 0.91 F1. Production: 0.84. Three weeks of panic.

WHY
Real
A real reason this matters in industry today.

SCALE
Today
A real reason this matters in industry today.

SPEED
Now
A real reason this matters in industry today.

IMPACT
Yours
A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1. Training-serving skew.
2. What a feature store does.
3. Feast · Tecton · SageMaker · Databricks.
4. DVC for dataset versioning.
5. Design a feature store for V-Align.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01
Training-serving skew
The silent killer

02
What a feature store does
Five duties

03
Feature store landscape 2026
Key concept of this lesson.

04
DVC — Git for data
Code versioning, extended

01

PART 1 OF 4

Training-serving skew

The silent killer

CONCEPT 1 OF 4

Training-serving skew

Training path: Jupyter notebook math.
Serving path: production microservice math.
Subtle differences compound; model's live accuracy drops.
Fix: SAME code computes BOTH.
CURE
One definition

Feature store guarantees the same code.

DEFINITION

Training-serving skew

The silent killer

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

What a feature store does

Single source of truth for definitions.
Serves both offline + online from same code.
Handles point-in-time joins.
Versions feature definitions.
Monitors feature freshness + quality.
MAGIC
Same code both ways

Skew disappears.

KEY POINTS

- Single source of truth for definitions
- Serves both offline + online from same code
- Handles point-in-time joins
- Versions feature definitions

CONCEPT 3 OF 4

Feature store landscape 2026

A core idea you'll use throughout this lesson.

DEFINITION

Feature store landscape 2026

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

DVC — Git for data

dvc init alongside git.
dvc add <file> → metadata in git, data in S3.
dvc push / pull — like git for binary.
Each commit pins a data version.
GUARANTEE
Reproduce any past run

git checkout + dvc pull.

KEY POINTS

- dvc init alongside git
- dvc add <file> → metadata in git, data in S3
- dvc push / pull — like git for binary
- Each commit pins a data version

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK · PAIR · SHARE

ACTIVITY

Pause and reflect

PROMPT

Apply the four concepts above to one real example from your own week.

HOW TO RUN IT

1.

2.

3.

4. Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1	*Where will you apply this in your next project?*
Q2	*What was the single biggest 'aha' from this lesson?*
Q3	*Which habit will you start using this week?*

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

KNOWLEDGE CHECK · QUESTION 1 OF 3

Point-in-time join prevents:

- **A** Speed
- **B** Future label leakage
- **C** Typos
- **D** Nothing

ANSWER · QUESTION 1 OF 3

Point-in-time join prevents:

CORRECT ANSWER

B

Future label leakage

WHY

Joins features AS-OF a specific timestamp

KNOWLEDGE CHECK · QUESTION 2 OF 3

Feast is:

A Managed

B Open-source feature store

C Model registry

D Vector DB

ANSWER · QUESTION 2 OF 3

Feast is:

B

CORRECT ANSWER

Open-source feature store

WHY

Open-source feature store

KNOWLEDGE CHECK · QUESTION 3 OF 3

DVC augments:

- **A** Docker
- **B** Git for data
- **C** K8s
- **D** Random

ANSWER · QUESTION 3 OF 3

DVC augments:

CORRECT ANSWER

B

Git for data

WHY

Git handles code; DVC handles data

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Training path: Jupyter notebook math.
Serving path: production microservice math.
Subtle differences compound; model's live accuracy drops.
Fix: SAME code computes BOTH.
CURE
One definition

Feature store guarantees the same code.

THE TAKEAWAY

The right tool, applied to the right problem, beats the loudest idea every time.

WHAT TO NOTICE

1. Training-serving skew
2. What a feature store does
3. Feature store landscape 2026
4. DVC — Git for data

APPLY IT → ASSIGNMENT

Define features once.

"Serve from that definition forever."

DELIVERABLES

- NEXT
- Lesson 6 — A/B testing ML.
- Document one decision you made because of this lesson.
- Document one decision you made because of this lesson.

WHAT 'GOOD' LOOKS LIKE

ONE THING TO REMEMBER

"

Knowing the right tool is half the battle. Doing the small thing today is the other half.

— AI Learning Hub

END OF LESSON 6

What you can now do

Each one of these is now part of your working vocabulary.

1. Pick the right tool — never the loudest one.
2. Practice the framework on a real example.
3. Watch for the three classic pitfalls.
4. Document your decision in one sentence.
5. Carry the discipline forward to the next lesson.